

The Effect of AI Writing on Governance: Evidence from a \$3.5 Billion DAO

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Motivation

- ▶ Text-based deliberation underpins financial governance:
SEC comment letters, proxy fights, analyst reports, Reddit
- ▶ It rests on one implicit assumption:

*Credible, contextually appropriate prose is costly to produce —
costly enough to screen out noise.*

- ▶ **Generative AI eliminates that cost**

Research question: Does online deliberation aggregate information before collective votes — and what happens to that information when producing credible text becomes costless?

This paper

Laboratory: the Arbitrum DAO — \$3.5B treasury, binding token votes, no boards or intermediaries. Proposals are debated on a public forum before each binding vote; every post and vote is recorded.

1. **Pre-AI, the forum aggregated information:** human post count predicts vote contestation ($r = -0.33$)
2. **The signal collapsed at the AI capability threshold:** post-shock $r = -0.10$; Chow $F = 15.05$ (GPT-4 Turbo), 8.38 (Claude 3)
3. **The mechanism is expected contamination, not realized volume:** AI was $\sim 5\%$ of posts at the structural break; the volume surge arrived 9–12 months *later*
4. **Bottom line:** the forum was worth **at least 555 words of research effort per delegate per proposal** — all destroyed, discretely, at the tipping point

Roadmap

1. The setting and the data

2. A model of forum deliberation — and why we need one
3. Main result: the signal collapse
4. Mechanism: expected contamination, not realized volume
5. Welfare: the forum in words of human effort

The Arbitrum DAO

Stakes

- ▶ $\approx$ \$13B worth of ARB tokens issued at launch (Mar 2023)
- ▶ $\approx$ \$3.5B worth (27.7% of supply) in the DAO treasury — spent only by token-weighted vote
- ▶ Economic model: grant treasury ARB to developers $\rightarrow$ apps that use ARB $\rightarrow$ demand raises ARB's value for everyone

Proposal lifecycle

- ▶ **Forum thread** — public discussion
- ▶ Snapshot temperature check
- ▶ Binding on-chain vote

Delegates (liquid democracy)

- ▶ Revocable delegation of voting power
- ▶ Top 10 delegates $\approx$ 30% of votes; a majority requires only **4 delegates**

Also in the ecosystem

- ▶ LobbyFi vote-rental market [▶ backup](#)

Data

- **Forum** (Discourse scrape): 17,553 posts, 1,101 topics, 2,452 users, Jan 2023–Mar 2026
- **AI detection** (Pangram API): *fraction_ai* per post; AI-written if > 0.70 . Scores are strongly bimodal — not sensitive to threshold ▶ [distribution](#)
- **Votes** (Snapshot): margin of victory = $\left| s_F / (s_F + s_A) - \frac{1}{2} \right| \times 2$
- **Matched sample**: **102 proposal–thread pairs** ($n_{\text{pre}} = 29$, $n_{\text{post}} = 73$)
- **Delegates**: wallet–username links for 62 of 218 posting delegates (hand-collected)

	Pre–Claude 3	Post–Claude 3	Change
Monthly posts	387	512	+32%
Monthly AI posts	8	143	+1,688%

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Why a model? Two purposes

Purpose 1: justify the empirical test.

- ▶ Why is the posts–margin correlation *the* measure of forum value?
- ▶ The model makes it an equilibrium object: **the forum aggregates information if and only if posting volume is negatively correlated with the margin of victory** (Prop. 1)

Purpose 2: enable welfare statements.

- ▶ A vanished correlation is not, by itself, a welfare statement
- ▶ The model prices the forum's information against participation costs, in calibratable units: **words of human writing effort**

A third implication: the model predicts the signal can collapse *discretely, at low AI volume* — exactly what the data show.

Model setup

(Lohmann 94; Austen-Smith 90; Feddersen–Pesendorfer 96; Dewatripont–Tirole 99)

Environment. A DAO evaluates proposal i of unknown quality $\theta_i \in \{H, L\}$, with prior $\Pr(\theta_i = H) = p_i$.

Contributors (a continuum, indexed j):

- ▶ Each holds a private signal $s_j \in \{h, l\}$ with precision $\Pr(s_j = \theta_i) = q > \frac{1}{2}$
- ▶ Posting the signal to the forum costs $c_p > 0$ per post

The delegate (the decision-maker):

- ▶ Votes For or Against; payoff $+1$ if the vote matches θ_i , -1 otherwise
 - ▶ relaxing alignment
- ▶ Observes *only* the forum; reading and interpreting one post costs $c_r > 0$
- ▶ Reading a genuine human post yields expected informational value $I_H > 0$ (in units of the voting payoff)

Equilibrium without AI

Contributors post only when the expected influence on the vote justifies the cost c_p . Both sides post in proportion to their support, so:

$$\mathbb{E}[\text{posts}_i] \propto p_i(1 - p_i) \quad \mathbb{E}[\text{margin}_i] \propto |2p_i - 1|$$

- ▶ Posting volume peaks at $p_i = \frac{1}{2}$ (contested proposals draw both sides in)
- ▶ The margin of victory is *smallest* at $p_i = \frac{1}{2}$
- ▶ The two are opposite monotone functions of contestedness $|p_i - \frac{1}{2}|$

Proposition 1 (Signal Content)

In the no-AI equilibrium, $\text{Cov}(\text{Posts}_i, \text{Margin}_i) < 0$. **An informative forum *must* show this negative correlation — this is our empirical test.**

The AI shock

At a point in time T , it becomes possible to emulate a genuine human post — carrying zero information about θ_i — at essentially zero cost ($c_p^{\text{AI}} \approx 0$).

Two objects now describe the forum:

- ▶ $\lambda \in [0, 1]$: the fraction of posts that *are* AI-generated
⇒ uninformative posts **in fact**
(a slow-moving realized share)
- ▶ $\pi \in [0, 1]$: the delegate's expectation that the next post read is AI-generated
⇒ uninformative posts **in expectation**
(the forward-looking expectation of λ)

Each object damages the forum through a different channel.

Two channels of damage

High λ : noise in fact

- ▶ Delegates read posts that carry no signal $\Rightarrow$ more incorrect votes
- ▶ Posting volume no longer tracks division $\Rightarrow$ the Prop.-1 correlation is diluted
- ▶ Damage is *gradual*: proportional to λ

High π : noise in expectation

- ▶ Value of reading a post:
 $(1 - \pi) I_H - c_r$
- ▶ For $\pi > \pi^* \equiv 1 - c_r/I_H$, reading stops paying: delegates abandon the forum *entirely*
- ▶ With no reader, writing genuine posts stops paying too: a **self-fulfilling collapse** (Akerlof 1970)

Proposition 2 (Unverifiability Threshold)

For $\pi > \pi^*$ the delegate reads nothing and the forum's *entire* informational value vanishes at once — even while most posts are still genuine (λ small).

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The posts–margin relationship breaks exactly once

Estimating equation (human = Pangram ≤ 0.70):

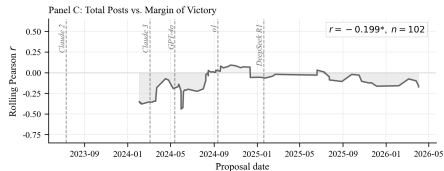
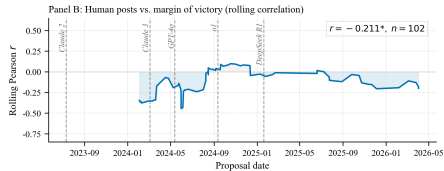
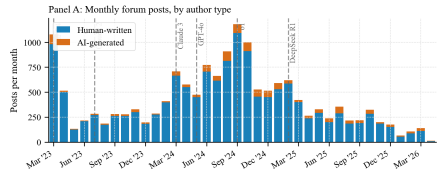
$$\text{Margin}_i = \alpha + \beta \text{HumanPosts}_i + \varepsilon_i$$

	(1) Full	(2) Pre–Claude 3	(3) Post–Claude 3
Human posts ($\hat{\beta}$)	–0.0015 (0.0012)	– 0.0026 ** (0.0011)	–0.0006 (0.0007)
N	102	29	73
Pearson r	–0.21	– 0.33	–0.10

Chow test of equal slopes: $F = 8.38$, $p = 0.0004$

- Pre-shock: an active thread is a sufficient statistic for community division — the forum *passes* the Prop.-1 test
- Post-shock: it *fails* — so by Prop. 1 this is a test of forum *value*, not just a correlation shift

AI Posting Volume, AI Model Release Dates and Rolling Correlations



Breaks track model capability — and precede the rise in AI volume

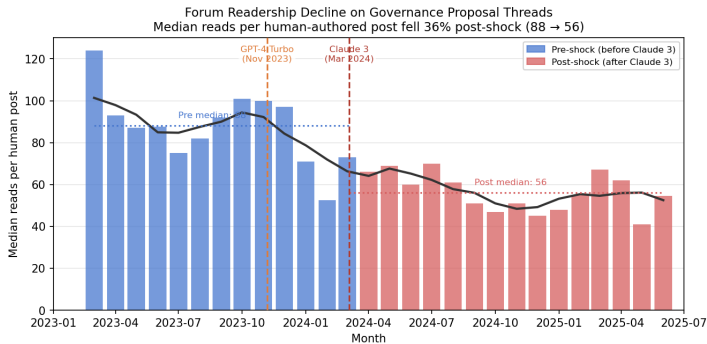
Shock	Pre-break		Post-break		F	Chow
	n	r	n	r		p
Claude 2 (Jul 23)	7	-0.17	95	-0.22	1.00	0.371
GPT-4 Turbo (Nov 23)	18	-0.34	84	-0.17	15.05	< 0.001
Claude 3 (Mar 24)	29	-0.33	73	-0.10	8.38	0.0004
GPT-4o (May 24)	36	-0.30	66	-0.06	5.97	0.004
o1 (Sep 24)	65	-0.20	37	-0.18	1.63	0.200
DeepSeek R1 (Jan 25)	78	-0.17	24	-0.38	0.67	0.513
QLR Sup- F	17.19 at Nov. 25, 2023 — 19 days after GPT-4 Turbo					

- Breaks cluster at the first models able to emulate human governance prose — not at later, better models ▶ [Turing-test evidence](#)
- **Timing puzzle:** AI $\approx$ 5% of posts at *both* breaks; the 15–20% surge came 9–12 months later
- The volume mechanism requires contamination *before* collapse — **rejected**. Consistent with the Proposition-2 tipping point ▶ [distraction checks](#)

Roadmap

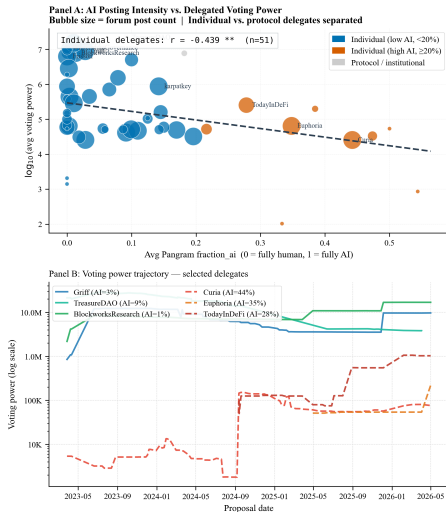
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Delegates stopped reading: direct behavioral evidence



- ▶ The forum platform records how many users open each post: the median falls 88 → 56 (−36%), **starting November 2023** — before any AI volume surge
- ▶ Not mechanical dilution: posts +122% would predict −55% under a fixed reading budget; aggregate reading *grew* ~40%
- ▶ Exit precedes evidence — delegates responded to the *prior*, as Prop. 2 predicts

Who adopts AI? Low-power delegates



- ▶ Cross-section ($n = 51$):
 $r = -0.44$ ($p = 0.001$)
- ▶ High-AI delegates hold $\approx \$70K$ worth of ARB; low-AI $\approx \$1.9M \Rightarrow 27\times$ gap
▶ delegate incentives
- ▶ AI posts carry no directional signal ($r = -0.009$ on pass/fail) ▶ content analysis
- ▶ $\Rightarrow$ unverifiability degraded the signal — not strategic manipulation by insiders
▶ panel evidence

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Welfare in the model: what the collapse destroys

Social welfare = expected vote correctness – participation costs:

$$W = p \Pr(\text{pass} \mid H) + (1 - p) \Pr(\text{reject} \mid L) - c_p \mathbb{E}[n]$$

Past the tipping point ($\pi > \pi^*$) the delegate votes at the prior, and

$$W^* - W^{\text{AI}} = \underbrace{4p(1-p)}_{\text{contestedness}} \cdot \underbrace{\left(q - \frac{1}{2}\right)^2}_{\text{signal quality}} > 0$$

- ▶ Loss is maximized at $p = \frac{1}{2}$: **AI destroys the most welfare on the most contested decisions** — where governance matters most
- ▶ Higher reading costs lower the tipping point ($\partial \pi^* / \partial c_r < 0$); more informative posts raise it ($\partial \pi^* / \partial I_H > 0$)
- ▶ The onset of indistinguishability is *first-order and discrete*; additional AI volume has no further marginal effect

Calibration: the forum, denominated in words of human effort

Object	Source	Value
c_p (write one post)	avg. pre-period human post length	237 words
$\bar{n}$ (human posts/proposal)	1,283 posts / 29 proposals	44.2
Writing cost per proposal	$\bar{n} \cdot c_p$	10,480 words
π^* (tipping threshold)	λ at QLR break, a lower bound ($\pi \geq \lambda$)	≥ 0.053
c_r (read one post)	$(1 - \pi^*) I_H$, normalizing $I_H = c_p$	≤ 224 words
Net reading surplus	$I_H - c_r$	≥ 12.56 words/post

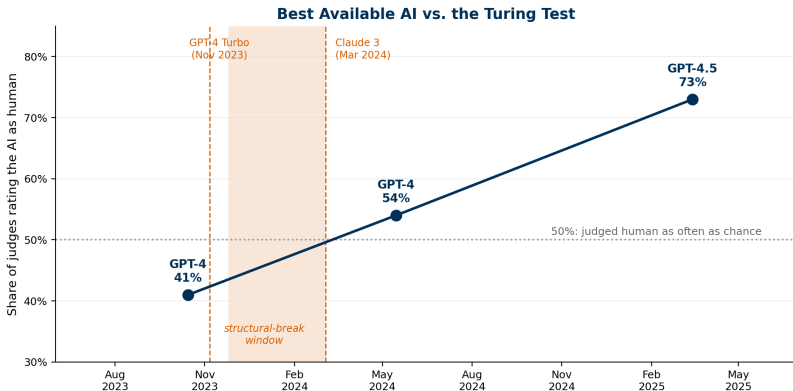
- Per delegate per proposal: $\bar{n} \times 12.56 \approx$ **at least 555 words** of research effort saved
- DAO-wide: at least $555K - 10,480$ words — positive for $K \geq 19$; at $K = 62$ delegates: **$\geq 24,000$ words per proposal** ($2.3\times$ the writing effort)
- Identification: $\pi^* = (I_H - c_r)/I_H$ — **the tipping point equals the per-post surplus share**. We observe only λ , and $\pi \geq \lambda$ while λ rises, so the collapse point bounds the margin from below: $\geq 5.3\%$ of gross value. All figures are conservative

Conclusion

Indistinguishable AI prose destroyed the forum — discretely, at 5% AI volume.

- ▶ The forum *did* aggregate information (Prop.-1 test passes pre-AI), generating at least 24,000 words of research-effort surplus per proposal
- ▶ The mechanism is **expected contamination, not realized volume**: expectations crossed the reading threshold before AI post volume did
- ▶ **Policy**: governance is underprovided (positive externalities), and the natural remedy is to subsidize participation — as Arbitrum does, paying delegates per posted rationale. **In the AI era this does worse than buy noise**: each subsidized zero-content post pushes expected contamination toward π^* — the subsidy can finance the collapse of the very forum it was meant to support ▶ DIP
- ▶ The same mechanism applies to shareholder meetings, peer review, and regulatory comment periods

Backup: AI crossed the human-indistinguishability line in the break window

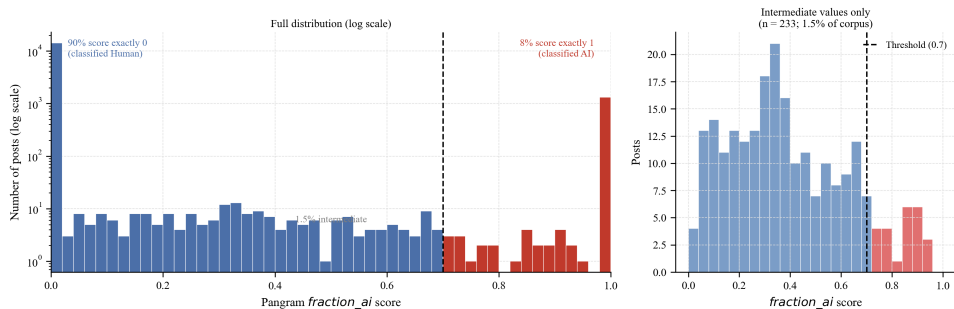


Interactive Turing tests, Jones & Bergen (UCSD): arXiv 2310.20216, 2405.08007, 2503.23674. Best-performing model/prompt per wave. Wave 3 is a three-party design (>50% = chosen over the actual human).

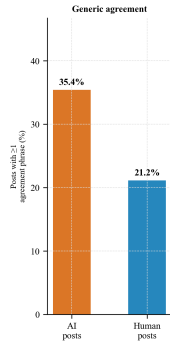
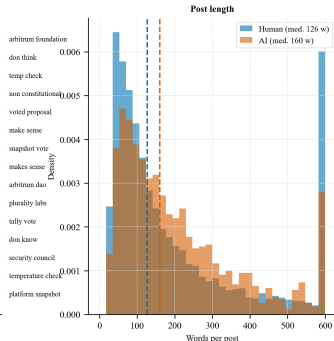
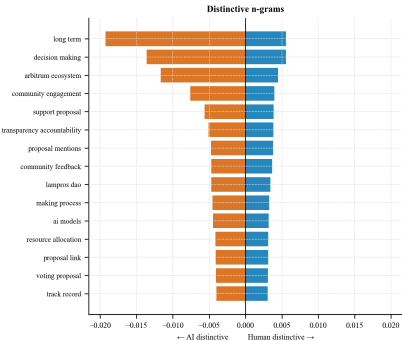
- ▶ Controlled interactive Turing tests (Jones & Bergen, UCSD; three waves, best model per wave). The 50% crossing falls inside the Nov 2023–Mar 2024 structural-break window
- ▶ Caveats: prompts improved across waves; conversational text, not governance prose

Backup: Pangram score distribution (bimodality)

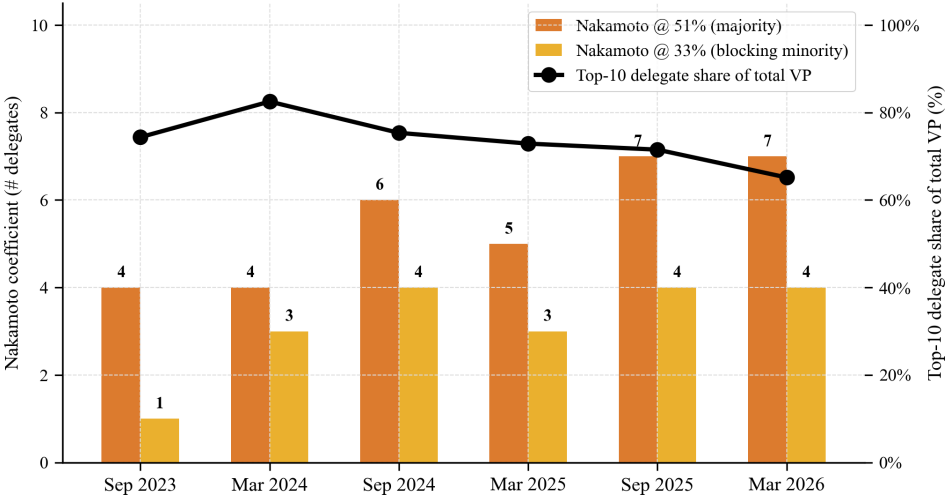
Distribution of Pangram AI-Detection Scores — Arbitrum DAO Forum Posts



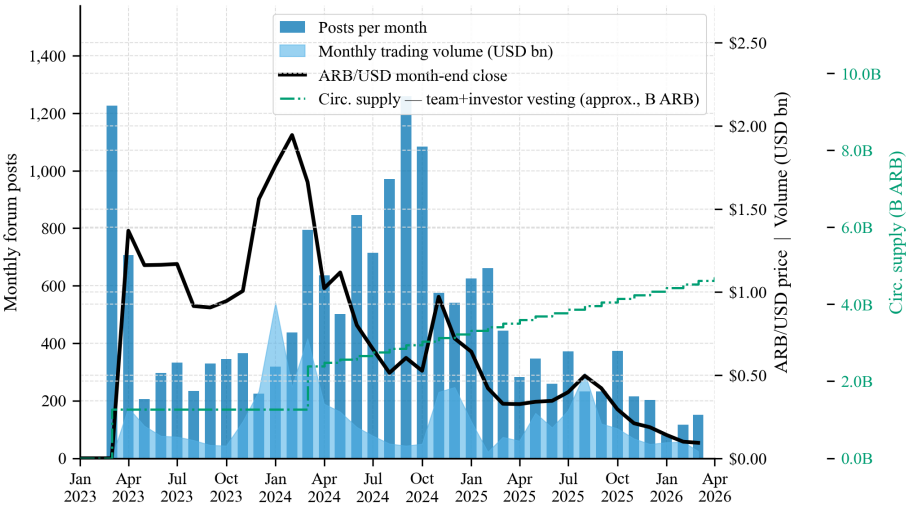
Backup: content analysis — AI vs. human vocabulary



Backup: voting power concentration



Backup: forum activity and token market

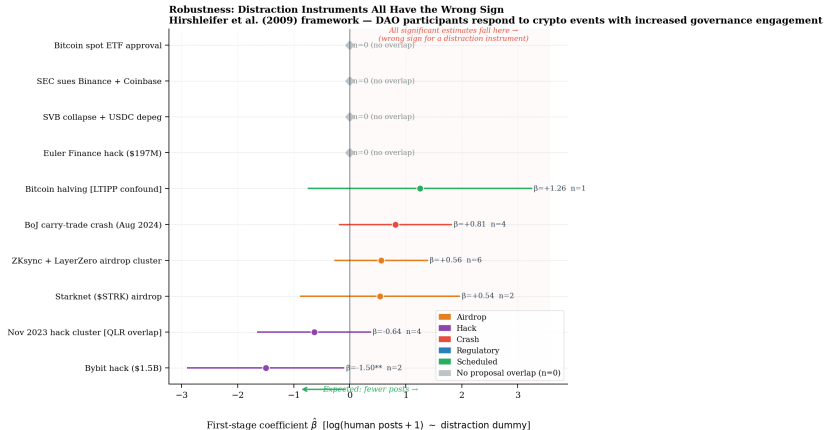


Backup: the distraction battery (every point estimate wrong-signed)

Candidate instrument	$\hat{\beta}$	t	Sign
ZKsync + LayerZero airdrops (May 24)	+0.696	1.69*	wrong
Starknet airdrop (Feb 24)	+0.714	0.94	wrong
BoJ rate shock (Aug 24)	+0.983	1.83*	wrong
Nov 23 hack cluster	+0.289	0.46	wrong
Concurrent proposal load	+0.168	1.38	wrong
ETH realized vol (monthly)	+35.5	4.18***	wrong
Bitcoin halving (Apr 24)	+0.149	0.76	wrong

- Dependent variable: log forum post count; proposal-clustered SEs
- DAO participants are core stakeholders: market chaos is a *complement* to governance engagement (contra Hirshleifer et al. 2009)

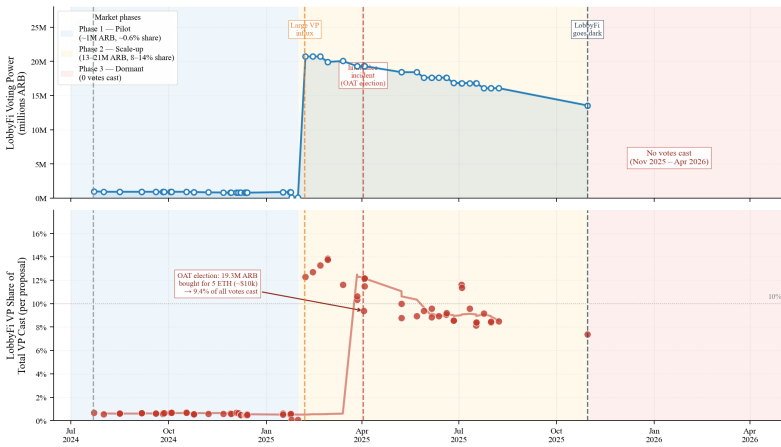
Backup: distraction first stages



Note: Instruments with n=0 treated proposals reflect no Arbitrum voting during the event window. Bitcoin halving (green, -sign) is confounded by the Arbitrum LTIPP grant batch (n=40 small applications in April 2024). Full sample: N=102 contested proposals, 2023–2026.

Backup: LobbyFi vote rental market

Figure A3: The Vote Rental Market — LobbyFi on Arbitrum DAO
Delegated Voting Power and Governance Share of the LobbyFi Proxy, July 2024 – April 2026
(Each dot = one Snapshot proposal vote; 78 votes total, 0 after October 2025)



Backup: the Delegate Incentive Program (DIP)

What it is (SEEDGov-administered; first program month March 2024):

- ▶ Monthly ARB payments to delegates scored on voting participation *and* publicly posting voting rationales (up to ~5,000 ARB/month in later versions)
- ▶ Eligibility: >50K ARB voting power (~175 delegates), >25% historical participation, KYC; 38 of 40 qualified in month one

Economics: governance participation has positive externalities, so a subsidy is sensible — but DIP subsidizes the *observable proxy* (posted text). With AI text at zero cost, the subsidy invites zero-content compliance prose — a concrete pecuniary motive for AI adoption by exactly the low-power delegates we identify. And it is worse than waste: each subsidized post raises expected contamination toward π^* , so **the subsidy can finance the discrete collapse of the forum it exists to support.**

- ▶ **Not a confounder for the break:** the QLR break (Nov 25, 2023) and the readership decline (Nov 2023) both *predate* the program — ruled out by timing, like the vesting cliff
- ▶ Plausibly amplifies the post-2024 volume surge and the posts-per-user increase

Backup: sketch — imperfectly aligned owners, endogenous delegation (WIP)

Ingredients. Holders with stakes w_i delegate to maximize token value; they observe delegates' *activity*, not their information. Delegate j 's payoff: pro-rata value of correct decisions ($\propto$ pivotality, rising in delegated power D_j) + delegation rents $R(D_j)$ (DIP, fees, reputation) – reading costs.

Key change: reading thresholds become delegate-specific, $\pi_j^* = 1 - c_r/l_j$ with l_j rising in stakes $\times$ pivotality; delegation flows respond to *observable activity* — which AI makes free.

If we solved this out, we would probably see:

- ▶ *Sorted, partial exit:* reading survives only at the top of the power distribution — aggregate readership falls but not to zero (consistent with the observed –36%)
- ▶ Post-shock, delegation flows *decouple from posting activity* (activity pools at zero cost) and concentrate toward incumbents $\Rightarrow$ voting power concentration rises
- ▶ AI adoption by low-power, rent-seeking delegates as an *equilibrium outcome* — the $27\times$ gap derived, not assumed

Our data could tell us: whether the activity $\rightarrow$ delegation elasticity falls post-break (delegate panel); whether concentration rises (Nakamoto series); whether small delegates herd on whales' votes (vote timestamps).

Backup: within-delegate panel (content, not volume)

	(1)	(2)
AI fraction (30-day)	0.055***	0.054***
Log posts (30-day)		0.001
Log voting power (lagged)	-0.047***	-0.047***
Delegate FE	Yes	Yes
<i>N</i> (delegate-periods)	1,832	1,832

- ▶ Within delegate: full AI adoption $\Rightarrow$ +5.5% voting power, swamped by mean reversion (-4.7%/period)
- ▶ Post *count* has no effect once content is controlled